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To the Editors of the International
Journal of Public Health

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Dear Editors,

We are pleased to submit our manuscript entitled “COVID-19 disrupted patterns of cause-specific mortality in Switzerland” for consideration in *the International Journal of Public Health*. Using high-quality national mortality data and a Bayesian multivariate Poisson model, we quantify cause-specific excess mortality during 2020–2021 and reveal how deaths from different causes are interdependent.

We analyzed weekly deaths in Switzerland from 2011–2021 by age, sex, region, and nine cause groups, accounting for seasonality, long-term trends, population changes, and cross-cause correlations. Cardiovascular deaths peaked during COVID-19 waves, while no cause showed sustained excess over the full period. In individuals ≥ 80 years, declines in non-COVID respiratory (-25%) and mental/neurological (-12%) deaths were largely offset by COVID-19 mortality, resulting in only a modest all-cause excess. We also identified strong pre-pandemic correlations between respiratory, cardiovascular and mental/neurological deaths ($\rho > 0.8$), and observed that excess mortality across these causes remained closely correlated during the pandemic.

Our framework formally models these cross-cause correlations and leverages the COVID-19 pandemic as a natural experiment to disentangle mechanisms that have historically overlapped. We identified three key drivers of cause-specific disruptions: misclassification of deaths, mortality displacement, and effects of control measures.

These findings provide insight into how pandemics reshape mortality beyond direct infection, highlighting the interplay between infectious diseases, chronic conditions, and public health measures. By clarifying these dynamics, our study contributes to a more accurate interpretation of disease burden, with direct relevance for public health decision-making, in line with the scope of the journal.

Thank you for your time and consideration. We look forward to your response.

Sincerely,

Dr. Anthony Hauser on behalf of all authors

